

Exponential Calibration Laws for Noncommuting Matrix-Polynomial and Linear-Phase MIMO FIR Filters

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Abstract

Scalar Chebyshev filtering treats every vector in a block Krylov iterate with the same polynomial, while simultaneously diagonalizable matrix coefficients amount to independent scalar filters after a fixed channel rotation. We study the larger class of Hermitian matrix-polynomial filters under tangential pass constraints. Already at fixed channel dimension $d = 3$, we construct rationally generated signatures with quantitative full-spark margin for which a symmetric noncommuting filter has stopband leakage $O(e^{-cN})$, whereas every exactly feasible pairwise-commuting symmetric filter is the identity. The signature margin and the admissible commuting calibration error are both $e^{-O(N)}$, not e^{-N^3} . This exponential scale is unavoidable: for arbitrary pass nodes and signatures in the same separated bands, an explicit scalar binomial-tail polynomial has both pass error and stopband leakage at most $e^{-2N/81}$. Thus constant-error separation is impossible, while the noncommuting construction achieves the correct exponential scale class. An affine cosine substitution gives the same robust law for fixed-latency reciprocal linear-phase MIMO FIR filters. We also give an exact rational five-tap certificate with stopband norm at most $25/32$ and prove that its advantage survives commuting calibration errors $\delta < 7/1920$. A second exact-arithmetic certificate is calibrated from a public triaxial pump-vibration data set: it gives leakage below 0.96 versus one for the exactly calibrated commuting class, while its maximum directional error on six held-out records is 0.00385. In a frequency-domain-decomposition test on the frozen held-out cospectra, filtering rotates the leading modal direction by at most 0.222° and changes its leading spectral ordinate by at most 2.4×10^{-5} relatively. Numerical programs test the mechanism and expose an ordinary graph-denoising setting in which scalar Chebyshev filtering is instead preferable. Tangential matrix interpolation, MIMO filtering, scalar two-band approximation and convex FIR design are not claimed as new; the contribution is the fixed-order commuting/noncommuting separation together with its two-sided calibration scale.

1 Problem and contribution

Let

$$P(x) = \sum_{k=0}^N C_k x^k, \quad C_k \in \mathbb{R}^{d \times d},$$

and let $I_- = [-1, 0]$ be a stopband. Given distinct pass points $\lambda_j > 0$ and nonzero channel signatures $v_j \in \mathbb{R}^d$, impose

$$P(\lambda_j)v_j = v_j, \quad 1 \leq j \leq M. \quad (1)$$

The leakage is

$$\rho(P) = \sup_{x \in I_-} \|P(x)\|_{\text{op}}. \quad (2)$$

The coefficients will always be real symmetric. They need not commute.

The comparator is the pairwise-commuting subclass. Such coefficients admit a common orthogonal eigenbasis, so a fixed channel rotation turns P into a diagonal collection of scalar

filters. The question is whether allowing the channel mixing to depend on polynomial degree can reduce (2) under the same pass constraints, and at what calibration-error scale that comparison remains meaningful. For unit signatures define

$$\gamma(v_1, \dots, v_M) = \min_{|J|=3} \sigma_{\min}([v_j]_{j \in J}).$$

Our answer is an exponential calibration law. The input signatures are full spark and have an explicit $e^{-O(N)}$ lower margin. This makes the commuting obstruction quantitative under $e^{-O(N)}$ residuals, while a separate scalar construction proves that no asymptotic advantage can survive calibration errors bounded below independently of N .

2 Main separation theorem

Theorem 2.1 (Robust exponential tangential separation). *There are absolute constants $C, c > 0$ and $N_0 \in \mathbb{N}$ such that for every $N \geq N_0$, with $d = 3$ and $M = N + 3$, one can choose distinct rational pass points $\lambda_1, \dots, \lambda_M \in [1, 7/2]$ and full-spark unit vectors $v_1, \dots, v_M \in \mathbb{R}^d$ satisfying*

$$\gamma(v_1, \dots, v_M) \geq \underline{\gamma}_N := \frac{2^{-39N}}{48(N+4)^6}, \quad (3)$$

with the following properties.

1. There is a degree- N polynomial P_N with real symmetric coefficient matrices satisfying (1) and

$$\rho(P_N) \leq Ce^{-cN}.$$

2. If a degree- N polynomial Q_N has pairwise-commuting real symmetric coefficient matrices and satisfies the same tangential constraints, then $Q_N(x) = I_d$ for every x . In particular, $\rho(Q_N) = 1$.

3. More generally, if such a commuting Q_N obeys

$$\max_j \|(Q_N(\lambda_j) - I)v_j\|_2 \leq \delta,$$

then, with $L_N = (24e/5)^N$,

$$\rho(Q_N) \geq 1 - \frac{\sqrt{3}L_N}{\underline{\gamma}_N} \delta. \quad (4)$$

Consequently $\rho(Q_N) \geq 3/4$ whenever $\delta \leq \underline{\gamma}_N / (4\sqrt{3}L_N)$.

For every $N \geq N_0$, the coefficient matrices of P_N cannot commute pairwise.

The admissible error in the third item is $e^{-O(N)}$. We split the proof into an obstruction, a small-leakage base construction, a quantitatively full-spark rational perturbation, and a matching scale obstruction.

2.1 The commuting obstruction

Lemma 2.2. *Let $M \geq d + N$, let the λ_j be distinct, and let the $v_j \in \mathbb{R}^d$ be full spark. If $Q(x) = \sum_{k=0}^N D_k x^k$ has pairwise-commuting real symmetric coefficients and $Q(\lambda_j)v_j = v_j$ for every j , then $Q = I_d$.*

Proof. The matrices D_k have a common orthonormal eigenbasis $u_1, \dots, u_d$. Write $q_r(x)$ for the scalar degree- N polynomial induced by $Q(x)$ on u_r . Taking the u_r component of the j th interpolation constraint gives

$$(q_r(\lambda_j) - 1)\langle u_r, v_j \rangle = 0.$$

A nonzero vector can be orthogonal to at most $d - 1$ of the full-spark targets: otherwise d targets would lie in the $(d - 1)$ -dimensional hyperplane $u_r^\perp$ and could not span $\mathbb{R}^d$. Thus at least $M - d + 1 \geq N + 1$ distinct values of λ_j are roots of $q_r - 1$. Consequently $q_r = 1$. This holds for every r , so $Q = I_d$. $\square$

2.2 A three-channel diagonal base filter

Put $M = N + 3$ and $S = \lceil M/3 \rceil$. Partition M nodes into three groups of sizes $s_g \in \{S - 1, S\}$, $g = 0, 1, 2$, and define

$$\lambda_{g,r} = 1 + \frac{5(3r + g)}{6S}, \quad 0 \leq r < s_g. \quad (5)$$

The nodes are distinct rationals in $[1, 7/2]$, their global separation is at least $5/(6S)$, and consecutive nodes within one group are separated by $h = 5/(2S)$.

Write T_m for the Chebyshev polynomial of the first kind, set $\phi(x) = 2x + 1$, and let $L_{g,r}$ be the Lagrange cardinal polynomial for the nodes in group g . With $m_g = N - s_g + 1$, define

$$q_g(x) = \sum_{r=0}^{s_g-1} L_{g,r}(x) \frac{T_{m_g}(\phi(x))}{T_{m_g}(\phi(\lambda_{g,r}))}, \quad P_0(x) = \text{diag}(q_0(x), q_1(x), q_2(x)). \quad (6)$$

Each q_g has degree at most N and equals one at every node in its group.

Lemma 2.3. *Let*

$$\eta_0 = \text{arcosh}(3), \quad \eta_1 = \text{arcosh}(8), \quad \Delta\eta = \eta_1 - \eta_0, \quad K = \frac{36e}{5},$$

and set

$$c = \frac{2\eta_0 - \log K}{3} > 0, \quad C_0 = 2 \exp\left(\frac{2}{3}(\log K + \eta_0)\right).$$

For every $N \geq 7$,

$$\sup_{x \in I_-} \|P_0(x)\|_{\text{op}} \leq C_0 e^{-cN}.$$

Moreover, at every pass node,

$$\|P_0(\lambda_j)\|_{\text{op}} \leq B_N := 2(4e)^{S-1} e^{N\Delta\eta}.$$

Proof. For $x \in I_-$, $|x - \lambda_{g,r}| \leq 9/2$. Equispacing within a group gives

$$\sum_{r=0}^{s_g-1} |L_{g,r}(x)| \leq \frac{(18S/5)^{s_g-1}}{(s_g - 1)!}.$$

Here we used $\sum_r [r!(s_g - 1 - r)!]^{-1} = 2^{s_g-1}/(s_g - 1)!$. For $N \geq 7$, $S \geq 4$, $s_g \geq S - 1$, and hence $S/(s_g - 1) \leq 2$. The bound $n! \geq (n/e)^n$ therefore gives

$$\sum_r |L_{g,r}(x)| \leq K^{s_g-1}. \quad (7)$$

Also, for $\lambda \geq 1$,

$$T_m(\phi(\lambda)) \geq \frac{1}{2} e^{m\eta_0}.$$

Combining this with (7) yields

$$|q_g(x)| \leq 2K^{s_g-1}e^{-(N-s_g+1)\eta_0}.$$

Since $s_g - 1 \leq S - 1 \leq (N + 2)/3$, the claimed $C_0 e^{-cN}$ follows. The constant c is positive because $e < 3$ gives $K < 108/5 < e^{2\eta_0} = (3 + \sqrt{8})^2$.

At another pass node the numerator distances in the Lagrange factors are at most $5/2$, so the same calculation gives $\sum_r |L_{g,r}(\lambda_j)| \leq (4e)^{s_g-1}$. The Chebyshev ratio is at most $2e^{N\Delta\eta}$, proving the pass-node bound. The operator norm of a diagonal polynomial is its largest entry. $\square$

2.3 An explicit full-spark perturbation

Index all pass nodes by $j = 1, \dots, M$, let $g(j)$ be the coordinate group of node j , and set $t_j = j/(M + 1)$. Define

$$w_j = (1, t_j, t_j^2)^\top, \quad \varepsilon_N = 2^{-13N}, \quad \tilde{v}_j = e_{g(j)} + \varepsilon_N w_j, \quad v_j = \frac{\tilde{v}_j}{\|\tilde{v}_j\|_2}. \quad (8)$$

Lemma 2.4 (Quantitative rational full spark). *The vectors in (8) are full spark and satisfy (3) for every $N \geq 7$.*

Proof. Fix any three indices and write

$$D(z) = \det(e_{g(j_1)} + zw_{j_1}, e_{g(j_2)} + zw_{j_2}, e_{g(j_3)} + zw_{j_3}) = \sum_{r=0}^3 d_r z^r.$$

The cubic coefficient is the nonzero Vandermonde determinant of the three w_j . Every d_r is rational with denominator dividing $(M + 1)^6$; therefore every nonzero coefficient has magnitude at least $(M + 1)^{-6}$. Expanding the determinant into its six signed products also gives $|d_r| \leq 18$.

Let r_0 be the first nonzero coefficient. For $N \geq 7$, $\varepsilon_N \leq [108(M + 1)^6]^{-1}$, and hence

$$\sum_{r>r_0} |d_r| \varepsilon_N^{r-r_0} \leq 18(\varepsilon_N + \varepsilon_N^2 + \varepsilon_N^3) \leq \frac{1}{2(M + 1)^6}.$$

It follows that

$$|D(\varepsilon_N)| \geq \frac{\varepsilon_N^{r_0}}{2(M + 1)^6} \geq \frac{\varepsilon_N^3}{2(M + 1)^6}.$$

Each unnormalized column has norm below two, so normalization loses at most a factor eight in the determinant. For a matrix with three unit columns the product of its two largest singular values is at most three. Its smallest singular value is consequently at least

$$\frac{\varepsilon_N^3}{48(M + 1)^6} = \frac{2^{-39N}}{48(N + 4)^6}.$$

The elementary inequality on ε_N holds at $N = 7$ and its left-to-right ratio decreases thereafter. $\square$

Lemma 2.5 (Stable symmetric interpolation). *For every $N \geq 7$, there is a real-symmetric coefficient polynomial P_N satisfying $P_N(\lambda_j)\tilde{v}_j = \tilde{v}_j$ and*

$$\sup_{x \in I_-} \|P_N(x) - P_0(x)\|_{\text{op}} \leq e^{-cN}.$$

Proof. Let $\mathcal{S}_N$ be the finite-dimensional space of degree-at-most- N polynomials with real symmetric coefficients, and define

$$L_\varepsilon : \mathcal{S}_N \longrightarrow (\mathbb{R}^d)^M, \quad L_\varepsilon(P) = (P(\lambda_j)(e_{g(j)} + \varepsilon w_j))_{j=1}^M.$$

At $\varepsilon = 0$, this map is onto. Indeed, for an off-diagonal entry $p_{rs} = p_{sr}$, arbitrary output data prescribe values at the union of nodes assigned to coordinates r and s . That union contains at most $2S \leq N + 1$ nodes for $N \geq 7$. Ordinary degree- N Lagrange interpolation is therefore onto on those nodes. A diagonal entry sees at most S nodes.

Equip the output space with the maximum Euclidean block norm and $\mathcal{S}_N$ with

$$\|P\|_X = \max \left\{ \sup_{x \in I_-} \|P(x)\|_{\text{op}}, \max_j \|P(\lambda_j)\|_{\text{op}} \right\}.$$

This also gives an explicit linear right inverse. To record its norm, let $y = (y_j)_{j=1}^M$ satisfy $\max_j \|y_j\|_2 \leq 1$. For $r \neq g$, define the entry $(R_N y)_{rg} = (R_N y)_{gr}$ by scalar Lagrange interpolation on the union of groups r and g : its value is $(y_j)_r$ at a node in group g and $(y_j)_g$ at a node in group r . A diagonal entry uses only its own group. These prescriptions are consistent because the groups are disjoint, and the resulting symmetric polynomial obeys $L_0 R_N y = y$.

For any such scalar problem, write its $t \leq 2S$ nodes in increasing order. Their separation $\delta = 5/(6S)$ implies that the denominator of the i th cardinal polynomial is at least

$$\delta^{t-1} i! (t-1-i)!.$$

On I_- and at every pass node, all numerator distances are at most $9/2$. Put $n = t - 1$. Since every scalar problem contains at least $S - 1$ nodes, $S/n \leq 2$ for $S \geq 4$. Consequently the sum of the absolute cardinal functions is bounded by

$$\left(\frac{27S}{5}\right)^{t-1} \sum_{i=0}^{t-1} \frac{1}{i!(t-1-i)!} = \left(\frac{27S}{5}\right)^{t-1} \frac{2^{t-1}}{(t-1)!} \leq \left(\frac{108e}{5}\right)^n.$$

The operator norm of a 3 by 3 matrix is at most three times its largest entry. Hence, with the norm $\|\cdot\|_X$ above,

$$\|R_N\|_{Y \rightarrow X} \leq 6 \left(\frac{108e}{5}\right)^{2S-1} =: \mathcal{R}_N. \quad (9)$$

Since $S = \lceil (N+3)/3 \rceil$, this explicit envelope satisfies $\mathcal{R}_N = \exp(O(N))$. Retaining the factorial is the step that improves the old superexponential perturbation scale.

Write $L_\varepsilon = L_0 + \varepsilon K_N$. Since $\|w_j\|_2 \leq \sqrt{3}$, $\|K_N\|_{X \rightarrow Y} \leq \sqrt{3}$. Hence

$$L_\varepsilon R_N = I + E_N, \quad \|E_N\| \leq \sqrt{3} \varepsilon \mathcal{R}_N.$$

For $\varepsilon = \varepsilon_N$ and $N \geq 7$, this is below $1/2$. Indeed, using $e < 11/4$, $e^{\Delta\eta} < 4$, $S \leq (N+5)/3$, and $108e/5 < 64$ gives

$$\mathcal{R}_N \leq 2^{4N+17}, \quad B_N \leq 2^{(10N+11)/3}. \quad (10)$$

Thus $\|E_N\| \leq 2^{18-9N} < 1/2$, so $(I + E_N)^{-1}$ exists and has norm at most two.

Since $L_0(P_0) = (e_{g(j)})_j$, the residual at the perturbed targets is

$$r_N = (\tilde{v}_j)_j - L_{\varepsilon_N}(P_0) = \varepsilon_N((I - P_0(\lambda_j))w_j)_j.$$

By Lemma 2.3,

$$\|r_N\|_Y \leq \varepsilon_N \sqrt{3}(1 + B_N).$$

Set

$$\Delta P_N = R_N(I + E_N)^{-1} r_N, \quad P_N = P_0 + \Delta P_N.$$

Then $L_{\varepsilon_N}(P_N) = (\tilde{v}_j)_j$ and

$$\|\Delta P_N\|_X \leq 2\sqrt{3} \mathcal{R}_N \varepsilon_N (1 + B_N) \leq e^{-cN}.$$

Indeed, $1 + B_N \leq 2B_N$, so (10) bounds the correction by $2^{(71-17N)/3}$. Since $c < \log 2$, the ratio of this envelope to e^{-cN} is at most $2^{(71-14N)/3} \leq 2^{-9}$ for $N \geq 7$. Every coefficient remains real symmetric because R_N maps into $\mathcal{S}_N$. $\square$

Lemma 2.6 (Quantitative commuting obstruction). *For the nodes and signatures above, let Q be a degree- N polynomial with pairwise-commuting real-symmetric coefficients and suppose*

$$\max_j \|(Q(\lambda_j) - I)v_j\|_2 \leq \delta.$$

Then (4) holds.

Proof. Simultaneously diagonalize the coefficients and fix a common unit eigenvector u , with scalar polynomial q . If three signatures all had $|u^\top v_j| < \underline{\gamma}_N/\sqrt{3}$, then the Euclidean norm of the three overlaps would be below $\underline{\gamma}_N$, contradicting (3). Thus at most two signatures have such a small overlap, and at $N + 1$ distinct nodes

$$|q(\lambda_j) - 1| \leq \frac{\sqrt{3}\delta}{\underline{\gamma}_N}.$$

For any selected $N + 1$ nodes, ordered increasingly, their separation is at least $5/(6S)$. The sum of the absolute degree- N Lagrange cardinal values at zero is at most

$$\left(\frac{42S}{5}\right)^N \frac{1}{N!} \leq \left(\frac{24e}{5}\right)^N = L_N,$$

where $S/N \leq 4/7$ for $N \geq 7$. Hence $|q(0) - 1| \leq \sqrt{3}L_N\delta/\underline{\gamma}_N$. Applying this to the common eigenbasis gives (4) because $0 \in I_-$. $\square$

Proof of Theorem 2.1. The normalized and unnormalized interpolation constraints are equivalent. Combine Lemmas 2.3, 2.4, and 2.5. The first conclusion holds with $C = C_0 + 1$ and the $c > 0$ of Lemma 2.3. The second conclusion is Lemma 2.2. The third is Lemma 2.6. Choose $N_0 \geq 7$ so that $Ce^{-cN_0} < 3/4$. Then the constructed coefficients cannot commute pairwise for $N \geq N_0$, and the robust comparison is strict at the displayed calibration radius. $\square$

2.4 Why the exponential calibration scale is unavoidable

The preceding radius is small, but its exponential order cannot be replaced by a constant. The obstruction is already scalar and is independent of the signature geometry.

Proposition 2.7 (Universal scalar noise barrier). *For every $N \geq 1$, arbitrary pass nodes $\lambda_j \in [1, 7/2]$, and arbitrary unit signatures v_j , there is a scalar degree- N polynomial s_N such that*

$$\sup_{x \in [-1, 0]} |s_N(x)| \leq e^{-2N/81}, \quad \max_j |s_N(\lambda_j) - 1| \leq e^{-2N/81}.$$

Consequently the commuting symmetric polynomial $s_N I$ has both leakage and maximum tangential calibration residual at most $e^{-2N/81}$.

Proof. Put $r(x) = 2(x + 1)/9$ and $k_N = \lceil N/3 \rceil$. Define the binomial-tail polynomial

$$s_N(x) = \sum_{k=k_N}^N \binom{N}{k} r(x)^k (1 - r(x))^{N-k}.$$

On the stopband $0 \leq r \leq 2/9$, while on the pass interval $4/9 \leq r \leq 1$. Hoeffding's inequality for a binomial random variable [11] gives

$$\Pr\{X/N \geq 1/3\} \leq e^{-2N/81} \quad (r \leq 2/9)$$

and

$$\Pr\{X/N < 1/3\} \leq e^{-2N/81} \quad (r \geq 4/9).$$

The ceiling in k_N only strengthens both deviations. These are precisely the two claimed bounds. $\square$

Thus exact feasibility yields a class separation, exponentially accurate feasibility retains it for the constructed signatures, and exponentially accurate scalar filters eventually erase it for every signature family. The sufficient and impossible tolerances do not have matching constants, but both have $-\log \delta = \Theta(N)$; an N -independent calibration floor is ruled out.

3 Block-Krylov meaning

The theorem has a direct interpretation for block polynomial filtering. Let $H = \sum_j \lambda_j q_j q_j^\top$ be real symmetric and let $B \in \mathbb{R}^{n \times d}$ be a starting block. For $P(x) = \sum_{k=0}^N C_k x^k$, define the right-coefficient block filter

$$\mathfrak{F}_P(H, B) = \sum_{k=0}^N H^k B C_k. \quad (11)$$

Its j th spectral row is

$$q_j^\top \mathfrak{F}_P(H, B) = (q_j^\top B) P(\lambda_j). \quad (12)$$

Proposition 3.1 (Pass preservation and stop contraction). *Let $\mathcal{T}$ and $\mathcal{S}$ index target and stop eigenpairs. Assume the rows $b_j^\top = q_j^\top B$ obey*

$$b_j^\top P(\lambda_j) = b_j^\top \quad (j \in \mathcal{T}), \quad \lambda_j \in I_- \quad (j \in \mathcal{S}).$$

Then the target spectral component is preserved exactly, and

$$\|Q_{\mathcal{S}}^\top \mathfrak{F}_P(H, B)\|_F \leq \rho(P) \|Q_{\mathcal{S}}^\top B\|_F,$$

where $Q_{\mathcal{S}}$ has columns q_j , $j \in \mathcal{S}$. If the C_k are pairwise-commuting symmetric matrices, one fixed orthogonal channel rotation reduces (11) to d independent scalar polynomial filters.

Proof. The pass statement follows from (12). For the stop part,

$$\|Q_{\mathcal{S}}^\top \mathfrak{F}_P(H, B)\|_F^2 = \sum_{j \in \mathcal{S}} \|b_j^\top P(\lambda_j)\|_2^2 \leq \rho(P)^2 \sum_{j \in \mathcal{S}} \|b_j\|_2^2.$$

For the final claim, simultaneously diagonalize all $C_k = U \Lambda_k U^\top$. Then the columns of BU are acted on by the scalar polynomials whose coefficients are the diagonal entries of the Λ_k . $\square$

Because every coefficient in Theorem 2.1 is symmetric, its column constraints transpose to the row constraints in Proposition 3.1. The theorem therefore gives block spectral instances with exponentially small stop contraction that no fixed-basis collection of scalar pass-preserving filters can reproduce. This is an approximation-theoretic separation; an end-to-end runtime advantage also depends on filter construction, recurrence stability, orthogonalization, and the cost model.

4 Reciprocal linear-phase MIMO FIR meaning

The polynomial separation is equivalently a fixed-order statement for a standard multichannel FIR class. Let

$$x(y) = \frac{9y + 5}{4}, \quad G_N(\omega) = P_N(x(\cos \omega)).$$

Because G_N is a real symmetric matrix cosine polynomial of degree N , it has a unique expansion

$$G_N(\omega) = A_N + 2 \sum_{r=1}^N A_{N-r} \cos(r\omega), \quad A_r = A_r^\top.$$

The causal response

$$H_N(e^{i\omega}) = \sum_{r=0}^{2N} A_r e^{-ir\omega} = e^{-iN\omega} G_N(\omega), \quad A_r = A_{2N-r}, \quad (13)$$

therefore has $2N + 1$ real symmetric palindromic taps and common group delay N .

Corollary 4.1 (Fixed-latency MIMO FIR separation). *For every $N \geq N_0$ there are $M = N + 3$ distinct frequencies $\omega_j \in [0, \arccos(-1/9)]$ and full-spark directions $v_j \in \mathbb{R}^3$ for which a causal response of the form (13) satisfies*

$$H_N(e^{i\omega_j})v_j = e^{-iN\omega_j}v_j, \quad \sup_{\omega \in [\arccos(-5/9), \pi]} \|H_N(e^{i\omega})\|_{\text{op}} \leq Ce^{-cN}.$$

Every response with the same real-symmetric palindromic tap structure whose taps commute pairwise and which satisfies the same calibrations is the pure delay $e^{-iN\omega}I$. Its high-frequency norm is one. If its phase-corrected calibration residual is at most δ , its high-frequency norm is at least the right-hand side of (4). Conversely, for arbitrary calibration directions and frequencies in the same pass range, there is a scalar palindromic response with both phase-corrected pass error and high-frequency norm at most $e^{-2N/81}$.

Proof. Take the data of Theorem 2.1 and choose $\cos \omega_j = (4\lambda_j - 5)/9$. The pass interval $[1, 7/2]$ maps into $[-1/9, 1]$, while the stop interval $[-1, 0]$ maps onto $[-1, -5/9]$. Equation (13) gives the calibration and transfers the norm bound unchanged. The affine substitution is invertible on coefficient spaces, as is the change between powers of $\cos \omega$ and the cosine basis. Thus the transformed coefficient matrices commute pairwise exactly when the original polynomial coefficients do. Palindromy makes the FIR taps an invertible relabelling and rescaling of the cosine coefficients. Applying Lemma 2.2 therefore forces the zero-phase response to be I . Lemma 2.6 transfers the residual bound, and Proposition 2.7 transfers the universal scalar response. $\square$

A fixed orthogonal channel rotation diagonalizes commuting symmetric taps, so the comparator in Corollary 4.1 is precisely a bank of independent scalar linear-phase filters in one frequency-independent channel basis. The result does not compare implementation cost, quantization robustness, or unconstrained MIMO filters, and the calibrated directions are constructed rather than taken from a deployed system.

5 Exact finite witnesses

5.1 A full-spark quadratic certificate

The asymptotic mechanism already occurs with rational data at degree two. Take

$$(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = \left(\frac{1}{2}, 1, \frac{3}{2}, \frac{7}{2}\right), \quad (v_1, v_2, v_3, v_4) = \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}\right).$$

Every pair of targets is linearly independent, hence the tuple is full spark in $\mathbb{R}^2$. Set $P(x) = C_0 + xC_1 + x^2C_2$, with

$$C_0 = \begin{pmatrix} 61/96 & 7/48 \\ 7/48 & 61/96 \end{pmatrix}, \quad C_1 = \begin{pmatrix} 7/24 & 7/48 \\ 7/48 & 77/96 \end{pmatrix}, \quad C_2 = \begin{pmatrix} -7/24 & -7/24 \\ -7/24 & -7/16 \end{pmatrix}.$$

Direct substitution verifies $P(\lambda_j)v_j = v_j$ for all four targets, and

$$[C_0, C_1] = \begin{pmatrix} 0 & 343/4608 \\ -343/4608 & 0 \end{pmatrix} \neq 0.$$

Proposition 5.1 (Exact quadratic gap). *The polynomial above obeys*

$$\sup_{x \in [-1, 0]} \|P(x)\|_{\text{op}} \leq \frac{25}{32} < 1.$$

Every pairwise-commuting real-symmetric quadratic polynomial satisfying the same four tangential constraints is identically I_2 and has leakage one.

Proof. For the upper bound, put $x = t - 1$ and express the leading principal minor and determinant of $(25/32)I_2 \pm P(t - 1)$ in the Bernstein basis on $t \in [0, 1]$. The leading-minor coefficients are

$$\left(\frac{35}{48}, \frac{7}{24}, \frac{7}{48}\right), \quad \left(\frac{5}{6}, \frac{61}{48}, \frac{17}{12}\right),$$

and the determinant coefficients are, respectively,

$$\begin{pmatrix} \frac{1421}{1536}, \frac{2597}{6144}, \frac{4655}{27648}, \frac{931}{18432}, 0 \end{pmatrix}, \\ \begin{pmatrix} \frac{1}{16}, \frac{1711}{3072}, \frac{3835}{3456}, \frac{3707}{2304}, \frac{1525}{768} \end{pmatrix}.$$

All are nonnegative and both leading minors are positive. Sylvester's criterion therefore gives $(25/32)I_2 \pm P(x) \succeq 0$ throughout the stopband, proving the norm bound. The zero final coefficient in the first determinant shows that the bound is attained at an endpoint rather than being a grid artifact.

For the commuting class, apply Lemma 2.2 with $M = 4 = d + N$. Every scalar common-eigenbasis entry has three distinct roots of $q - 1$ and is identically one. $\square$

The gap is stable under imperfect calibration. Write $\hat{v}_j = v_j / \|v_j\|_2$.

Proposition 5.2 (Robust commuting obstruction). *Let Q be a pairwise-commuting real-symmetric quadratic polynomial satisfying*

$$\|Q(\lambda_j)\hat{v}_j - \hat{v}_j\|_2 \leq \delta, \quad j = 1, \dots, 4.$$

Then

$$\sup_{x \in [-1, 0]} \|Q(x)\|_{\text{op}} \geq 1 - 60\delta.$$

In particular, if $\delta < 7/1920$, its leakage is strictly larger than the certified 25/32 leakage of the noncommuting witness.

Proof. For any two distinct normalized targets a, b in the tuple, $|a^\top b|^2 \leq 9/10$. Hence their two-column smallest singular value obeys

$$\sigma_{\min}^2 = 1 - |a^\top b| \geq \frac{1 - |a^\top b|^2}{2} \geq \frac{1}{20} > \frac{1}{25}.$$

For every unit vector u , at least one member of any pair therefore has $|u^\top \hat{v}_j| \geq 1/(5\sqrt{2})$. Thus at most one target has smaller overlap. Simultaneously diagonalize Q and fix a common unit

eigenvector u , with scalar quadratic entry q . Projecting the residual bound shows that at three distinct pass nodes

$$|q(\lambda_j) - 1| \leq 5\sqrt{2}\delta.$$

For each of the four possible three-node subsets of $\{1/2, 1, 3/2, 7/2\}$, the sum of the absolute Lagrange weights at $x = 0$ is at most eight. Hence $|q(0) - 1| \leq 40\sqrt{2}\delta < 60\delta$. This holds for both common eigenvectors, so $\|Q(0)\|_{\text{op}} \geq 1 - 60\delta$. Finally, $1 - 25/32 = 7/32$ gives the stated threshold. $\square$

The exact-arithmetic replay `verification/verify_quadratic_filter_witness.py` checks all constraints, target minors, commutators, Bernstein coefficients, and the rational robust-calibration constants without a numerical solver.

5.2 An exact five-tap MIMO FIR certificate

The quadratic witness has an exact causal FIR realization. Substitute $x = (9\cos\omega + 5)/4$ and define the palindromic taps

$$B_0 = B_4 = \begin{pmatrix} -189/512 & -189/512 \\ -189/512 & -567/1024 \end{pmatrix},$$

$$B_1 = B_3 = \begin{pmatrix} -63/128 & -21/32 \\ -21/32 & -21/64 \end{pmatrix}, \quad B_2 = \begin{pmatrix} -149/768 & -665/768 \\ -665/768 & -235/1536 \end{pmatrix}.$$

Their response obeys the exact identity

$$\sum_{r=0}^4 B_r e^{-ir\omega} = e^{-2i\omega} (B_2 + 2B_1 \cos\omega + 2B_0 \cos 2\omega) = e^{-2i\omega} P\left(\frac{9\cos\omega + 5}{4}\right).$$

Consequently it preserves the four directions, up to their common delay phase, at cosine values $-1/3, -1/9, 1/9, 1$ and has operator norm at most $25/32$ on $\omega \in [\arccos(-5/9), \pi]$. Every commuting symmetric palindromic five-tap response meeting the same four calibrations is the pure two-sample delay and has norm one. Moreover $[B_0, B_1] \neq 0$. All statements, including the tap conversion, are checked by the same exact-arithmetic replay. The same change of variables transfers Proposition 5.2: if a commuting five-tap comparator obeys $\|e^{2i\omega_j} H(e^{i\omega_j}) \hat{v}_j - \hat{v}_j\|_2 \leq \delta$, its high-frequency norm is at least $1 - 60\delta$.

5.3 An affine two-constraint precursor

For degree one and $d = 2$, take pass data

$$\lambda_1 = \frac{1}{2}, \quad v_1 = (1, 0)^\top, \quad \lambda_2 = 1, \quad v_2 = (1, 1)^\top,$$

and $P(x) = A + xB$ with

$$A = \begin{pmatrix} 4/5 & 1/5 \\ 1/5 & 3/10 \end{pmatrix}, \quad B = \begin{pmatrix} 2/5 & -2/5 \\ -2/5 & 9/10 \end{pmatrix}.$$

The constraints hold exactly and

$$[A, B] = \begin{pmatrix} 0 & -1/10 \\ 1/10 & 0 \end{pmatrix}.$$

Convexity of $x \mapsto \|A + xB\|_{\text{op}}$ puts its maximum over I_- at an endpoint. Direct diagonalization gives

$$\rho(P) = \max\{\|A\|_{\text{op}}, \|A - B\|_{\text{op}}\} = \frac{1 + \sqrt{61}}{10} < 0.882.$$

If A and B commute, their common eigenbasis contains a direction with nonzero overlap with both nonorthogonal, nonparallel targets. Its scalar affine entry equals one at two points and hence everywhere. The commuting leakage is at least one.

6 Numerical stress test

The repository script `research/route_c_scaling_pilot.py` solves a grid SDP with symmetric coefficient variables. For degree 3, dimension 9, 12 perturbed targets and seed 7, the results are:

Perturbation	SDP objective	Validation maximum	Commutator norm
0.001	0.2983490	0.2983682	0.466925
0.010	0.3199793	0.3200281	0.391179
0.030	0.3977121	0.3977588	0.708604

The optimization uses 81 stopband points and validation uses 4001 independent points. Clarabel labels these solves `optimal_inaccurate`; they are diagnostics, not proof. The exact witnesses and Theorem 2.1 do not depend on solver output.

6.1 Controlled block-iteration benchmark

The script `research/route_c_block_krylov_benchmark.py` embeds the quadratic certificate in a 256-dimensional Hermitian eigenproblem. The four pass eigenvalues are $1/2, 1, 3/2, 2$, the other 252 eigenvalues fill $[-1, 0]$, and the initial stop/pass Frobenius ratio is five. We repeatedly apply the degree-two matrix filter and compare it with

$$q(x) = \frac{8x^2 + 8x + 1}{7},$$

the scalar degree-two Chebyshev stopband filter normalized only at $x = 1/2$. Let $\sin \theta$ denote the largest principal-angle sine from the filtered block to the clean two-dimensional block defined by the four prescribed pass rows. The scalar pass error below is minimized over one global rescaling, so it does not merely report scalar amplitude growth.

Iterations	Matrix stop/pass	Matrix $\sin \theta$	Scalar pass error	Scalar $\sin \theta$
0	5.0000	0.9918	0.0000	0.9918
1	2.2207	0.9638	0.7518	0.7847
5	0.3752	0.5622	0.9195	0.9802
10	0.0760	0.1385	0.9199	0.9819
20	0.00434	0.00810	0.9199	0.9820

The matrix pass error is below 1.2×10^{-10} through ten iterations and is 7.8×10^{-4} at iteration 20 because floating-point errors are amplified outside the preserved directions; exact arithmetic preserves them. The scalar filter has stopband maximum $1/7$ and initially suppresses contamination much faster, but its pass multipliers are $(1, 17/7, 31/7, 127/7)$; iteration consequently rotates the clean block away from the prescribed one. The fair commuting comparator under all four exact tangential constraints is the identity and remains at stop/pass ratio 5 and $\sin \theta = 0.9918$. This controlled construction demonstrates both the separation and its recurrence-conditioning limitation, not a runtime advantage on a natural problem.

6.2 A measured triaxial calibration and exact continuum certificate

We next use VBL-VA001, an open lab-scale pump-vibration data set with triaxial acceleration records sampled at 20 kHz [2, 12]. The task is frequency-domain-decomposition (FDD) pre-processing, not fault classification. FDD extracts mode-shape estimates as leading singular vectors of cross-spectral matrices at spectral peaks [6]; filter-induced perturbations are therefore a scientifically relevant failure mode, and filtering artifacts can corrupt modal estimates [21]. We selected twelve normal-condition records by the archive prefixes `normal_000` and `normal_001`,

using all six records under the first prefix for fitting and retaining all six under the second prefix. The file names, CRC-32 and SHA-256 digests were frozen before optimization.

Each record was polyphase-decimated by eight, then divided into length-1024 Hann windows with 50% overlap. We averaged the real 3 by 3 cospectral matrix and, using the training split only, selected the five largest-trace peaks between 20 and 1000 Hz subject to an eight-bin separation. Their leading cospectral eigenvectors are the channel signatures. The frequency node is $x = \cos(2\pi f/2500)$. Table 1 reports the six-decimal rationalization used by the exact certificate; every displayed component and node differs from its measured value by less than 5×10^{-7} .

Lemma 6.1 (Modal-component invariance). *Let the cospectral matrix at a modal frequency have the decomposition $G_j = \lambda_j v_j v_j^\top + R_j$, where $\lambda_j \geq 0$, $\|v_j\|_2 = 1$, and $R_j \succeq 0$. If a real-symmetric response P obeys $P(x_j)v_j = v_j$, then the reciprocal palindromic FIR response $H(e^{i\omega}) = e^{-iN\omega} P(\cos \omega)$ obeys*

$$H(e^{i\omega_j})G_jH(e^{i\omega_j})^* = \lambda_j v_j v_j^\top + P(x_j)R_jP(x_j).$$

Thus the rank-one modal component, its channel shape, and its amplitude are preserved exactly, up to one common N -sample delay, while the residual cospectrum is filtered.

Proof. The scalar phase cancels in HG_jH^* , and symmetry gives $P(x_j)v_j v_j^\top P(x_j) = v_j v_j^\top$. $\square$

f (Hz)	x	v_x	v_y	v_z
100.0977	0.968522	-0.054531	0.996129	-0.068944
200.1953	0.876070	0.980644	0.167125	0.102012
744.6289	-0.296151	0.152303	0.986844	-0.054250
844.7266	-0.524590	0.869001	-0.350335	0.349432
944.8242	-0.720003	0.606651	0.716228	-0.344951

Table 1: Training frequencies and rationalized triaxial signatures from VBL-VA001.

Proposition 6.2 (Data-grounded rational certificate). *For the five rational nodes and signatures in Table 1, there is a real-symmetric quadratic polynomial P satisfying $P(x_j)v_j = v_j$ exactly and*

$$[C_k, C_\ell] \neq 0 \quad \text{for some } k, \ell, \quad \sup_{-1 \leq x \leq -929/1000} \|P(x)\|_{\text{op}} < \frac{24}{25}.$$

Every pairwise-commuting real-symmetric quadratic satisfying the same five calibrations is identically I_3 and has leakage one.

Proof. All ten 3 by 3 target determinants are nonzero in exact rational arithmetic; the smallest normalized absolute determinant is 0.009489456048.... Lemma 2.2, with $d = 3$, $N = 2$ and $M = 5$, therefore proves the commuting conclusion.

For the upper bound, fix the (1, 1), (1, 2) and (2, 2) entries of C_2 to -0.316171 , -0.537470 and -0.909040 , respectively, and solve the remaining 15 rational linear equations for the other symmetric entries. The exact replay verifies the calibrations and a nonzero commutator. On the 101-point rational grid of $[-1, -929/1000]$, Sylvester’s criterion gives

$$(191/200)^2 I_3 - P(x_i)^2 \succ 0.$$

The same replay proves $\sup_{|x| \leq 1} \|P'(x)\|_{\text{op}} \leq 5.740500963$. The grid spacing is $71/100000$, hence every point is within half a grid spacing of some x_i and

$$\|P(x)\|_{\text{op}} < \frac{191}{200} + 5.740500963 \frac{71}{200000} < 0.957038 < \frac{24}{25}.$$

Every comparison in this paragraph is performed on integers and rational fractions; the printed decimals only abbreviate upward-valid bounds. $\square$

The held-out signatures lie between 0.129° and 0.460° from their training counterparts. Without refitting, the rational witness has maximum held-out residual 0.0038491 and dense-grid leakage 0.94870. The numerical comparators in Table 2 use the unrounded measured nodes and signatures. A general nonsymmetric MIMO polynomial attains much lower training leakage, but its held-out residual is 0.1045 and its largest coefficient Frobenius norm is 36.06, compared with 2.623 for the rational symmetric witness. This single split suggests a reciprocity–robustness tradeoff; it does not establish a statistical generalization theorem.

For a task-level replay, we froze the full 3 by 3 training and held-out cospectral matrices at the five peaks, applied $P(x_j)G_jP(x_j)$ on the held-out split, and recomputed the FDD leading eigenpair. Across the five frequencies, the angle between the unfiltered and filtered leading directions is at most 0.2217° ; the leading eigenvalue ratio differs from one by at most 2.40×10^{-5} . These figures are outputs of the offline replay, not consequences of the exact certificate. Lemma 6.1 explains the specification: exact calibration preserves an ideal modal component, while the held-out calculation measures the effect of residual components and train–test drift.

Filter	Stop leakage	Fit tolerance	Held-out residual
Rational symmetric witness	< 0.96000	exact	0.003849
Symmetric SDP optimum	0.94867	exact	0.003849
General nonsymmetric SDP	0.14514	exact	0.10447
Scalar quadratic optimum	0.97520	0.01	0.01000
Best of 64 fixed bases	0.96966	0.01	0.01011
Exact commuting class	1	exact	0

Table 2: Measured-calibration controls. The 64-basis row is a seeded stress test, not a global optimization certificate over orthogonal bases.

Thus the measured instance closes both the purely constructed-data objection and the objection that tangential preservation has no task meaning: its calibration and validation use disjoint sensor records, FDD supplies the mode-shape objective, and the exact 0.96-versus-1 gap is not solver-dependent. It does not show improved fault diagnosis, damping estimation, hardware cost or deployment performance; those claims are outside the present experiment.

6.3 A failed graph-denoising application gate

We also tested the standard MIMO graph-filter form $Y = \sum_{k=0}^2 S^k X C_k$ on two smooth fields over an 18 by 23 grid graph, using four low-frequency full-spark row signatures. The symmetric MIMO SDP collapsed to the identity (validated stop norm 0.99999992), while the one-point-normalized scalar Chebyshev filter had stop norm $1/17$. After two iterations its best globally rescaled pass error was only 0.00524 and its desired-subspace sine was 0.0292, compared with 0.997 for the symmetric MIMO solution. A nonsymmetric MIMO fit eventually preserved the exact geometry, but was inferior at short iteration counts. Thus ordinary graph denoising does not justify the exact tangential specification, and no such application claim is made. We retain this negative result to prevent selecting an example merely because it matches the theorem’s notation.

7 Relation to established work and scope of the claim

Tangential matrix-polynomial interpolation is classical [4, 7]. Structured matrix-polynomial interpolation with Hermitian, positive-semidefinite and related symmetries is also established [1]; that work treats full matrix values and lists the tangential structured variant as a future problem. Matrix-valued Nevanlinna–Pick theory handles supremum-norm analytic interpolation, including complexity constraints [5, 3]. Kuroiwa and Lindquist treat the bitangential problem

with explicit McMillan-degree constraints [15]. Most directly, Stefanovski and Georgijević prove that real stable rational matrices can meet bitangential constraints while having arbitrarily small spectral norm on a proper frequency region [18]; their degree grows and they do not compare commuting and noncommuting Hermitian polynomial coefficients. Thus small regional norm under matrix tangential constraints is not itself our contribution. Tangential interpolation is also a standard bridge between MIMO model reduction and Krylov projection [8]; modern block rational approximation includes matrix-valued weights, noisy-data comparisons, and tangential interpolation mechanisms [10]. Recent iterative MIMO schemes even select low-rank tangential data using maximum-error criteria [13]. Their maximum-error step selects the next interpolation frequency, whereas their proved monotonicity concerns a weighted H_2 error; monotone H_∞ bounds are stated as future work. Gallivan et al.’s main result instead gives a generically unique minimal-McMillan-degree rational interpolant. Neither result provides a fixed-degree Hermitian-polynomial minimax separation from the simultaneously diagonalizable subclass. Minimax polynomial filtering and SDP formulations have a long numerical-linear-algebra history [19], and matrix-polynomial spaces arise naturally in block Krylov methods [17]. General MIMO graph filters already use matrix taps in precisely the form $\sum_k S^k X C_k$ [9], while scalar FIR minimax design by convex optimization is classical [20]. Kootsookos already formulates fixed-length MIMO FIR approximation of matrix transfer functions in the uniform H_∞ norm, proves global error lower bounds, and gives MIMO and linear-phase design algorithms [14]. More closely, Ljungars and Fu optimize the operator-norm error of matrix-valued multi-channel linear-phase FIR filters on a frequency grid by SDP [16]. Their formulation has neither directional pass interpolation nor a commuting/simultaneously-diagonalizable comparator. Accordingly, fixed-length MIMO H_∞ approximation, MIMO linear phase, operator-norm minimax design, and SDP are not claimed as new.

The candidate novelty is therefore narrower than norm-constrained or degree-constrained tangential interpolation or FIR design alone. Under fixed ordinary degree—equivalently fixed reciprocal FIR latency—a three-channel Hermitian-coefficient tangential minimax problem exhibits an exponential separation between fully coupled coefficients and the full fixed-basis scalar subclass. The new quantitative result also identifies the calibration scale: an exponentially small residual is sufficient for the constructed separation, whereas a universal scalar construction proves that an exponentially small residual can erase every such separation. We have not found this two-sided law in the sources above.

The complete texts of two close restricted papers—Fuhrmann (2010) and Stefanovski–Georgijević (2016)—could not be obtained. Publisher portals, repository copies, metadata APIs, and independent network routes were checked. Their accessible statements already rule out broader novelty claims, but absence of a theorem from inaccessible text is not evidence of priority. The claim is therefore limited to the displayed theorem and to the literature actually inspected. The FDD lemma and held-out replay support a preprocessing interpretation; they do not constitute a deployment or diagnostic study.

Reproducibility. The source and research ledger are in the project repository. The repository links the three-channel theorem arithmetic replay, the exact rational replay, the block-iteration benchmark, its machine-readable output, the deliberately negative graph-filter pilot, the VBL-VA001 provenance manifest, its exact continuum replay, the offline baseline replay, and its machine-readable output, the clause-by-clause priority matrix, and the artifact manifest with the PDF hash.

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